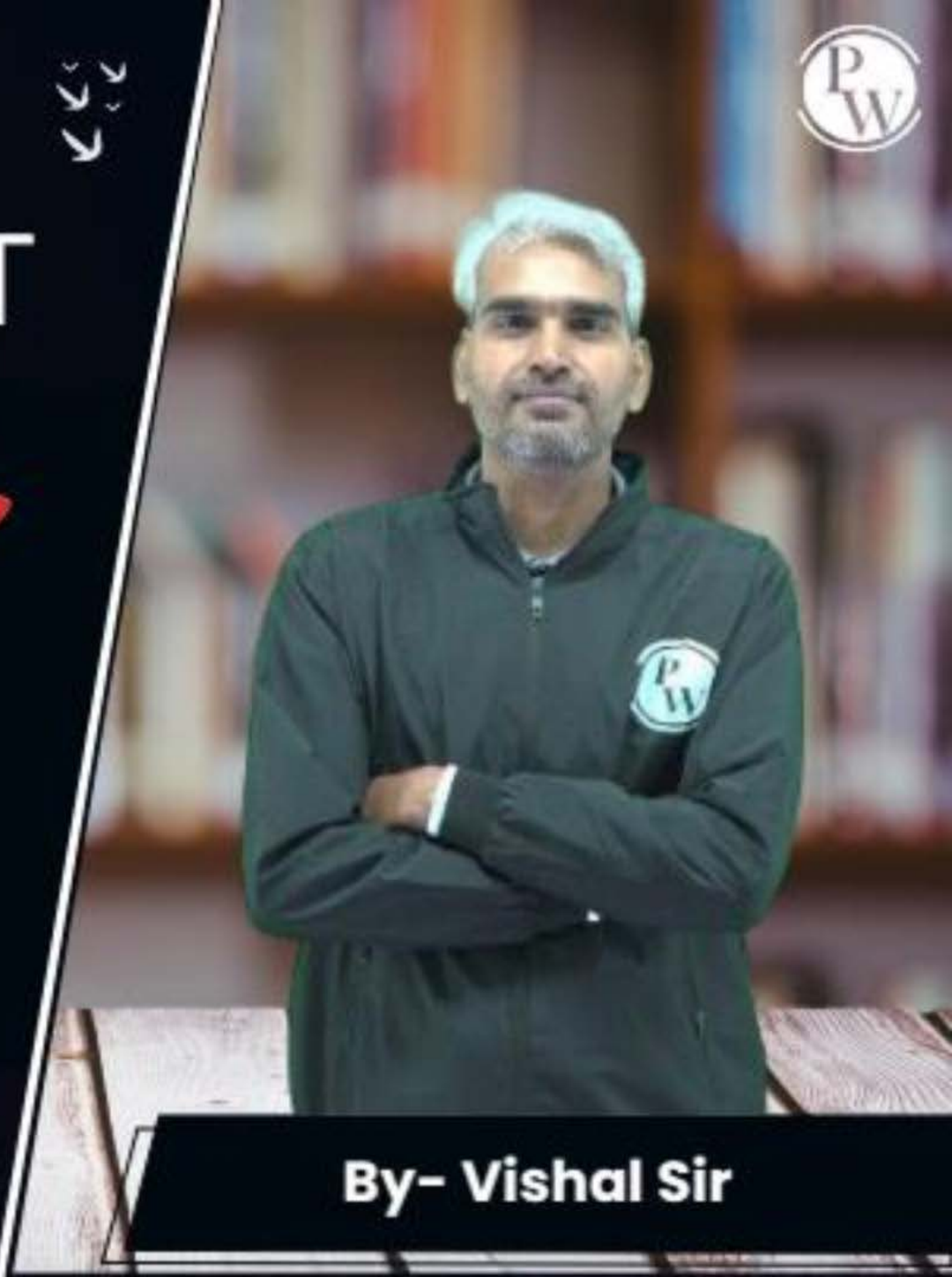
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



Topic Bi-partite graph

Topic Complete bi-partite graph

Topic Sum of degree theorem

Topic Degree sequence

Slide

Topics to be Covered



Topic — Quick revision

Topic — Havel Hakimi's result

Slide

Today's Topic

H.W

#Q. The number of edges in a k - regular graph with n vertices is

$\therefore k|V| = 2|E|$

$k * n = 2 * |E|$

$$|E| = \frac{nk}{2}$$

H.W.

|E| = 16

2-regular

#Q. A non-directed graph contains 16 edges and all vertices are of degree 2. Then the number of vertices in G is 16

$k|V| = 2|E|$

$2 * |V| = 2 * 16$

$|V| = 16$

H.W.

$|E| = 25$

#Q. G is undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is ~~16~~.

'deg' is at least '3'

$\therefore 3|V| \leq 2|E|$

$3 * |V| \leq 2 * 25$

$|V| \leq \frac{50}{3}$

$|V| \leq 16.66$ $\therefore$ $V_{max} = 16$



Topic : Havel Hakimi's Result

Used to check whether given degree sequence is a graphic or not

"Next Slide"

Havel Hakimi's Algorithm

Consider the following degree sequence in non-increasing order

Seq 1: $\{S, t_1, t_2, t_3, \ldots, t_S, d_1, d_2, \ldots d_K\}$ { $S, t_1, t_2, \ldots d_1, d_2, \ldots$ etc. are degrees of different vertices }

To reduce the size of the problem, delete the 1st degree from deg. sequence {i.e, delete 'S'}, and subtract '1' from next 'S' degrees in the degree sequence. And obtain a new sequence (let Seq 2)

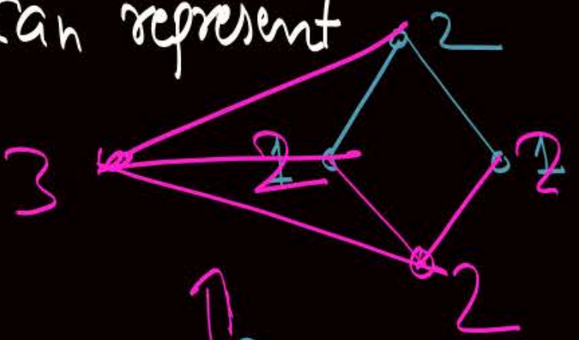
Seq 2: $\{(t_1-1), (t_2-1), (t_3-1), \ldots, (t_S-1), d_1, d_2, \ldots d_K\}$

"Seq 1" Can represent a simple non-directed graph if and only if "Seq 2" Can represent a simple non-directed graph

If we are not able to visalize the graph w.r.t. "Seq 2" as well, then rearrage the "Seq 2" in non-increasing order, and repeat the process to obtain "Seq 3"

{ "We can keep repeating until we are not able to visualize the graph }

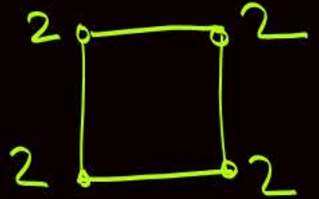
Q:- Check whether the following degree sequence can represent a simple non-directed graph or not?

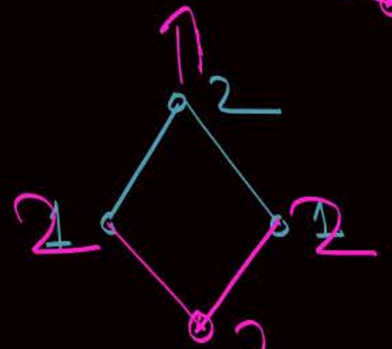


Seq 1: { ~~3~~, 3, 3, 3, 2 }

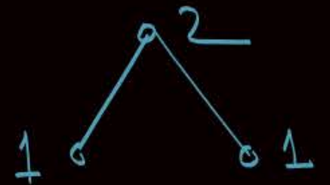
-1

Seq 2: { ~~2~~, 2, 2, 2, 2 }

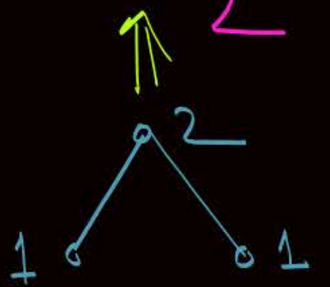




Seq 3: { 1 1 2 }



Rearranged Seq 3: { ~~2~~, 1 1 }



Seq 4: { 0 0 }





#Q. Which of the following degree sequences represent a simple non directed graph?

S1 = {6, 6, 6, 6, 4,3,3,0}

S2 = {6, 5, 5, 4, 3, 3, 2, 2,2}

A Only S1

B Only S2

C Both S1 and S2

D Neither S1 nor S2

$S_1 = \{6, 6, 6, 6, 4, 3, 3, 0\}$

Seq 1 = $\{\cancel{6}, 6, 6, 6, 4, 3, 3, 0\}$

Seq 2 = $\{\cancel{5}, 5, 5, 3, 2, 2, 0\}$

Seq 3 = $\{\cancel{4}, 4, 2, 1, 1, 0\}$

Seq 4 = $\{3, 1, 0, 0, 0\}$



Final sequence can not represent any simple - non-directed graph.

∴ 'S_1' is not a graphic

$S_2 = \{6, 5, 5, 4, 3, 3, 2, 2, 2\}$

Seq 1 : $\{6, 5, 5, 4, 3, 3, 2, 2, 2\}$

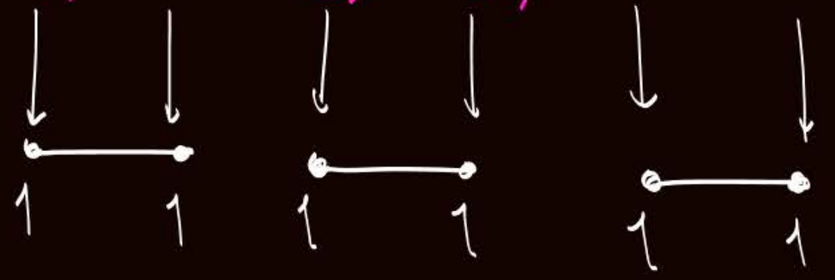
Seq 2 : $\{4, 4, 3, 2, 2, 1, 2, 2\}$

↳ Rearranged Seq 2 : $\{4, 4, 3, 2, 2, 2, 2, 1\}$

Seq 3 : $\{3, 2, 1, 1, 2, 2, 1\}$

↳ Rearranged Seq 3 : $\{3, 2, 2, 2, 1, 1, 1\}$

Seq 4 : $\{1, 1, 1, 1, 1, 1\}$



Seq "4" represent a simple non-directed graph, ∴ "S_2" will also represent a simple non-directed graph

#Q. Which of the following degree sequences represent a simple non directed graph?

S1 = {6, 6, 6, 6, 4,3,3,0}

S2 = {6, 5, 5, 4, 3, 3, 2, 2,2}

A Only S1

B Only S2

C Both S1 and S2

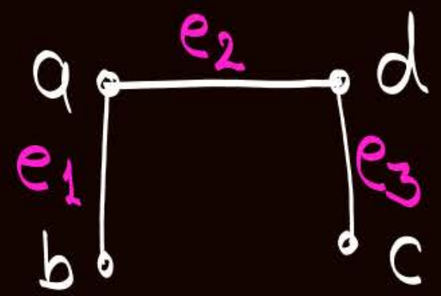
D Neither S1 nor S2

Complement of a graph

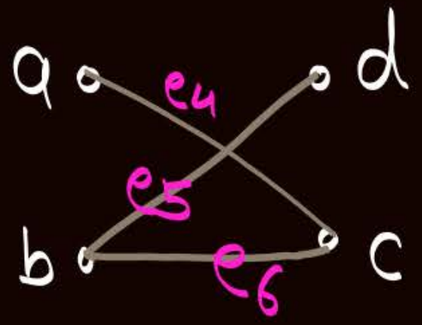
Complement can be defined for a graph G, if and only if 'G' is a simple graph

Let G be a simple graph with 'n' vertices, then complement of graph G can be denoted by '$\overline{G}$'. and $\overline{G}$ is also a simple graph with same set of vertices as of graph G, but an edge is present in $\overline{G}$ if and only if that edge is not present in G.

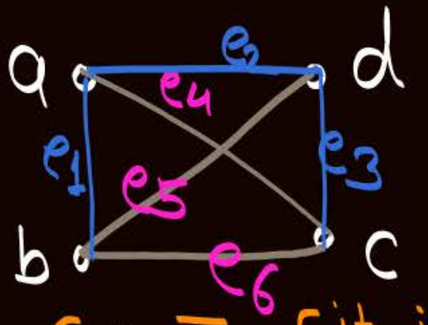
eg:



U



≡



$G \cup \bar{G}$ {it is a Complete graph}

G

$G = (V, E_1)$

$V = \{a, b, c, d\}$

$E_1 = \{e_1, e_2, e_3\}$

$\bar{G}$

$\bar{G} = (V, E_2)$

$V = \{a, b, c, d\}$

$E_2 = \{e_4, e_5, e_6\}$

$G \cup \bar{G} = (V \cup V, E_1 \cup E_2)$

$V \cup V = V = \{a, b, c, d\}$

$E_1 \cup E_2 = \{e_1, e_2, e_3, e_4, e_5, e_6\}$

Note: If G is a simple graph with 'n' vertices, then

$$G \cup \overline{G} = K_n$$

We know,

$$|V(G)| = |V(\overline{G})|$$

&

$$|E(G)| + |E(\overline{G})| = |E(K_n)|$$

$$|E(G)| + |E(\overline{G})| = \frac{n(n-1)}{2}$$

Q: Let G be a simple graph with 'n' vertices and '21' edges, if number of Edges in complement of graph G is '24' then find the no. of Vertices in graph G.

$$|E(G)| + |E(\bar{G})| = \frac{n(n-1)}{2}$$

$$21 + 24 = \frac{n(n-1)}{2}$$

$$n(n-1) = 90$$

$\hookrightarrow n = -9$ or $n = 10$

Number of Vertices can not be in negative.

$\therefore \boxed{n = 10}$

Note:- Let 'G' be a simple graph with 'n' vertices and following degree sequence,

Deg. Seq. of $G = \{x_1, x_2, x_3, \ldots\ldots, x_n\}$,

then

Degree Seq. of $\overline{G} = \{(n-1)-x_1, (n-1)-x_2, (n-1)-x_3, \ldots\ldots, (n-1)-x_n\}$

Note :-

Let G be a simple graph and $\bar{G}$ is the complement of graph G, then at least one of G or $\bar{G}$ is definitely connected

- Both G & $\bar{G}$ may be connected but both G & $\bar{G}$ can never be disconnected.



2 mins Summary

Topic Quick revision

Topic Havel Hakimi's result

Topic Complement of a graph

Slide

THANK - YOU